

Roll No.....

ODD END SEMESTER EXAMINATION DECEMBER - 2024

Name of the Program: **B. Tech. Petroleum Engineering**

Semester: **III**

Name of the Course: **General Geology**

Course Code: **TPE 302**

Time: **3 Hours**

Maximum Marks: **100**

Note:

- (i) All questions are compulsory.
- (ii) Answer any **two** sub questions among **a, b & c** in each main question.
- (iii) Total marks for each main question are **twenty**.
- (iv) Each question carries **10 Marks**

		(20 marks)
Q1		
(a)	What is the significance of minerals in Earth's crust, and how are they utilized in various industries?	CO1 CO2
(b)	How do weathering and erosion compare? Discuss the different processes that cause weathering in rocks.	
(c)	Explain the rock cycle and describe how rocks are formed, transformed, and recycled through its stages.	
Q2		(20 marks)
(a)	Illustrate the internal constitution of the Earth, and explain the characteristics of each of its layers.	CO1 CO2
(b)	Differentiate between i) Streak and Colour ii) Lava and Magma	
(c)	Define plate tectonics and explain the different kinds of plate boundaries that shape Earth's surface.	
Q3		(20 marks)
(a)	Discuss the process of formation of igneous rock.	CO2 CO3
(b)	Describe the classification system and the significance of sedimentary rocks.	
(c)	Describe the typical characteristics of the three basic types of rocks.	
Q4		(20 marks)
(a)	Define earthquake? Explore the factors that lead to its occurrence. Provide a brief explanation of the different types of seismic waves produced during an earthquake.	CO4 CO5
(b)	Define the terms 'focus' and 'epicenter' in relation to earthquakes. What measures should be taken to ensure safety when constructing buildings in earthquake-prone regions?	
(c)	What is meant by volcanism? Describe the various products produced during a volcanic eruption.	
Q5		(20 marks)
(a)	Explain the formation of folds and outline the different types, using clear sketches for better understanding	CO5 CO6
(b)	What are faults in geology? Explain their various types with the help of neat diagrams.	
(c)	Explain the concept of a joint in geology. What are the different types of joints that can be found?	

